

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

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ABSTRACT

Spatial overlap and mean class contrast need not imply target-specific region responses. We audit CCI’s maximum-target-drop region in frozen CLIP, using separate prompts for selection and evaluation. Across COCO and VOC with two checkpoints, 41.16–64.78% of locality/mean-foil screen-passers fail a worst-foil criterion. With original screening memberships retained, excluding annotated-present foils leaves 39.77–63.64% failing. Exclusion reduces failure beyond exact cardinality-matched random removal by 0.86–0.92 percentage points on COCO; VOC intervals include zero. Positive-target filtering retains high failure. Among screened failures, unrestricted eight-region oracles permit sign repair in 6.25–7.84% on COCO and 27.40–31.29% on VOC. A target-drop tolerance of $\epsilon = .02$ reduces feasible repair to 0.16–0.98%; relaxing it increases opportunity with setting-dependent target and locality changes.

Index Terms— CLIP, explanation auditing, class specificity, constrained region selection

1 INTRODUCTION

CLIP [1] region explanations use intervention-induced image–text score changes. Cluster-based Concept Importance (CCI) [2] groups patches and measures their contribution; we audit its maximum-target-drop region, CCI-top1. A positive mean-foil contrast can average away a strong competitor: masking a region may reduce the target score more than the average non-target score, yet reduce one non-target score even more. Worst-foil specificity tests this strongest competitor, complementing spatial overlap and mean contrast.

COCOA [3] attributes representation similarity to a reference corpus relative to foil samples. CASE [4] tests class sensitivity and constructs contrastive saliency maps. Our audit examines CCI-selected regions under held-out prompts, testing whether locality and positive mean-foil contrast imply a positive worst-foil margin across non-target dataset classes. It combines fixed-population annotation exclusion, cardinality-matched foil controls, and exact candidate enumeration to identify the sources of failure and limits on sign repair.

Related CLIP attribution [5, 6, 7], class contrasts [8], and caption foils [9] provide complementary perspectives. Worst-foil (WF) reranking probes the available candidate and budget space with CLIP fixed; IDEA [10] instead adapts CLIP for few-shot classification.

2 AUDIT AND REPAIR OPPORTUNITY

For checkpoint m , the pre-softmax score is $s_m(x, c) = \tau_m \langle z_m(x), e_m(c) \rangle$, with unit-norm embeddings and learned $\tau_m = \exp(\ell_m)$. Region R has intervention response

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c). \quad (1)$$

For target category t , selection uses “a photo of a {category}”:

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t). \quad (2)$$

Held-out margins and screens. Evaluation uses “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. For unit-norm text embeddings $e_h(c)$, let $H = 3$, $n_c = \|H^{-1} \sum_h e_h(c)\|_2$, and $\bar{d}_c = H^{-1} \sum_h d_h(c)/n_c$. With $\mathcal{F}_t$ the non-target dataset categories, the aggregate worst-foil margin is

$$P(R, t) = \bar{d}_t - \max_{f \in \mathcal{F}_t} \bar{d}_f. \quad (3)$$

This equals using normalized mean text embeddings; class-specific n_c can change rankings. Tables report raw target drop $H^{-1} \sum_h d_h(t)$, whereas the positive-target screen uses $\bar{d}_t$. Selection and evaluation use separate templates.

For CCI-top1, define $A = \{\text{bbox} \geq .5, \bar{d}_t - \text{mean}_f \bar{d}_f > 0\}$ and $F = \{P < 0\}$. We use .5 as the locality threshold. To distinguish positive mean-foil contrast from positive target contribution, we also audit $A^+ = A \cap \{\bar{d}_t > 0\}$. The primary sign-repair population is $B = A \cap F$, fixed before reranking. Throughout, “failure” denotes $P < 0$, and “sign repair” denotes a transition from $P < 0$ to $P \geq 0$.

Target-constrained diagnostic probe. Define

$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_t} d_R^{\text{sel}}(f), \quad (4)$$

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

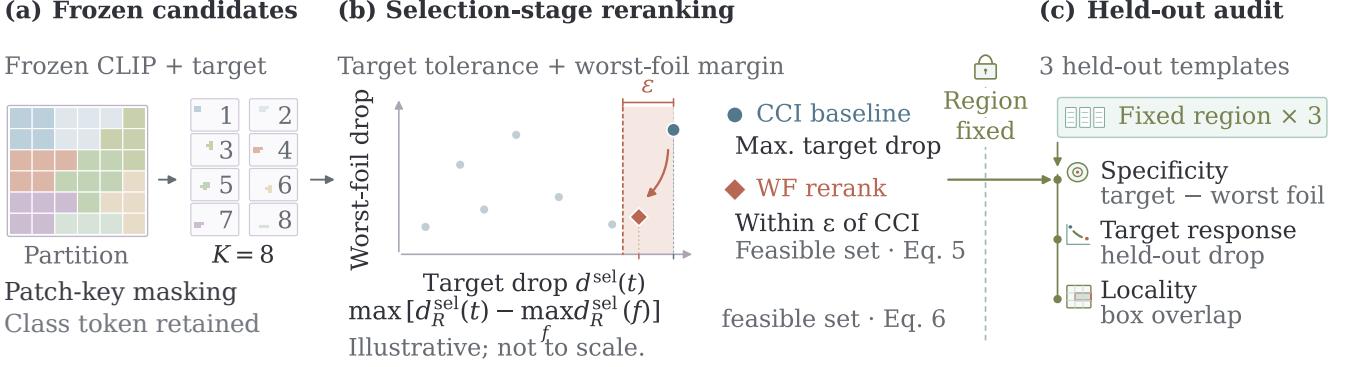


Figure 1: Selection and held-out audit. A region is fixed using selection-stage responses before three-template evaluation of specificity, target response, and locality. Schematic responses illustrate the selection rule.

and worst-foil (WF) reranking

$$R_{\text{WF}} = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

The baseline is feasible, so any switch loses at most ϵ of *selection-stage* target drop. We measure its held-out effects separately. Mean and Max-.1 instead maximize $d_t^{\text{sel}} - \text{mean}_f d_f^{\text{sel}}$ and $d_t^{\text{sel}} - .1 \max_f d_f^{\text{sel}}$ over the same set. The reference budget is $\epsilon = .02$ logit units. A post-hoc .01/.02/.05/.10/.20 sweep measures budget sensitivity within each checkpoint; unrestricted oracles search the full candidate set.

Exact diagnostic oracles. For the screened-failure population B , the feasible sign-repair count is

$$N_O(B, \epsilon) = \sum_{i \in B} \mathbf{1} \left\{ \max_{R \in \mathcal{R}_\epsilon(i)} P_i(R) \geq 0 \right\}. \quad (7)$$

The unrestricted oracle searches all K candidates. A sign pass means $P_i(R) \geq 0$, without locality or positive-target constraints. Failures split into: no passing candidate (C_0); passing candidates but *none feasible* (C_1); a feasible pass missed by WF (C_2); and WF sign repair (r). Thus $|B| = C_0 + C_1 + C_2 + r$ and $N_O = C_2 + r$. We additionally count repairs satisfying $J(R) = \{P_i(R) \geq 0, A(R), \bar{d}_t(R) > 0\}$; $A(R)$ applies the same full-foil screen to the new region, without changing original B . Oracles quantify held-out opportunity within the candidate set.

Fixed-population foil controls. For image i , let $\mathcal{F}_i^{\text{abs}}$ remove all annotated-present categories from the non-target foils. The protocol holds R_{CCI} and full-foil screen A fixed; $\mathcal{F}_i^{\text{abs}}$ tests $\bar{d}_t - \max_{f \in \mathcal{F}_i^{\text{abs}}} \bar{d}_f < 0$. Reducing the foil set weakly decreases failure, so we also match its cardinality. For N_i full foils, $k_i = |\mathcal{F}_i^{\text{abs}}|$, and m_i foils exceeding the target, a uniform size- k_i subset without replacement fails with probability

$$q_i = 1 - \frac{\binom{N_i - m_i}{k_i}}{\binom{N_i}{k_i}}. \quad (8)$$

With zero numerator when $k_i > N_i - m_i$, we compare $\Delta_A = |A|^{-1} \sum_{i \in A} (\mathbf{1}\{\mathcal{F}_i^{\text{abs}}\} - q_i)$. Restricted sets are nonempty, and crowd annotations are excluded.

3 PROTOCOL AND REPRODUCIBILITY

COCO val2014 [11] and VOC2007 test [12] provide 27,708/1,943 eligible images, each evaluated with OpenAI CLIP B/16 and B/32 (59,302 image–model records). The target is the largest instance’s category, using annotation area on COCO and box area on VOC. We require another category and 2–80% target area; COCO crowd annotations are excluded. Full foils are the other 79/19 classes.

K-means clusters L2-normalized projected patch features: $K = 8$, three initializations, 50 iterations, seed $1701 + i$ (sorted IDs shuffled with seed 1701 define i). Intervention $\mathcal{I}$ adds $-\infty$ to attention logits at selected patch-key columns before softmax in every block/head, retaining the class-token column. Ties use first argmax; MPS features are cast to float32. BBox precision is the fraction of selected patch centers in target-class box unions, mapped by $224 / \min(W, H)$ and center-crop offsets [13].

Float32 candidate analyses use stored responses. Foil controls regenerate full-class responses: IDs, CCI labels, feasible sizes, and full-foil states match; compared summaries differ by less than 10^{-5} . Original A, A^+ memberships are retained. These checks establish agreement of the recorded summaries; spatial reproducibility is discussed in Section 5.

Intervals use 10,000 image bootstraps and percentile 2.5/97.5 limits, pointwise without multiplicity correction. Paired quantities are differenced within image before re-sampling. Foil comparisons resample within fixed A/A^+ ; repair-cost intervals condition on actual repairs. Exact q_i is primary; 100 seeded draws check it.

4 RESULTS

Table 1: Fixed original A : full-foil, annotation-absent, and exact cardinality-matched random failure rates (%). Δ_A is absent minus random in percentage points, with paired 95% CI.

Setting	Full	Absent	Random	Δ_A
COCO B/16	64.37	63.17	64.09	-0.92 [-1.08,-0.77]
COCO B/32	64.78	63.64	64.50	-0.86 [-1.01,-0.72]
VOC B/16	42.27	40.94	41.46	-0.52 [-1.16,+0.06]
VOC B/32	41.16	39.77	40.32	-0.55 [-1.19,+0.04]

Failure persists after annotation exclusion. The fixed screen A contains 17,726, 17,847, 1,209, and 1,227 images in table order. Exclusion lowers failure by only 1.14–1.39 percentage points of A ; 96.63–98.24% of original failures B remain (Table 2). Table 1 separates this change from foil-count reduction: COCO exclusion has a larger effect than random removal of the same number of categories, whereas VOC Δ_A intervals include zero. The residual failure extends beyond the contribution of explicitly annotated co-occurring classes.

Positive-target and scene-count checks. The mismatch persists under positive-target filtering. Among A^+ , full-foil failure is 62.64–63.79% on COCO and 40.74–41.40% on VOC. After annotation exclusion, it remains 61.38–62.61% and 39.34–40.05%, respectively. Matched-random comparisons retain the COCO/VOC interval pattern. Even scenes with exactly two annotated categories have COCO annotation-absent failure of 63.39/64.95%.

Table 2: Exact local decomposition of *screened failures* $B = A \cap F$ at $\epsilon = .02$. Counts in C_0, C_1, C_2, r are mutually exclusive; $C_2 + r$ is the feasible-oracle count.

Setting	$ B $	C_0	C_1	C_2	r
COCO B/16	11,410	10,697	695	0	18
COCO B/32	11,561	10,655	875	1	30
VOC B/16	511	371	135	0	5
VOC B/32	505	347	155	1	2

Candidate and budget bottlenecks differ by dataset. No candidate passes for 92.16–93.75% of COCO screened failures B , versus 68.71–72.60% on VOC (Table 2). Unrestricted sign-repair capacity is thus 6.25–7.84% of COCO B and 27.40–31.29% of VOC B (Figure 2). The .02 budget excludes 96.43–98.10% of these opportunities, leaving feasible capacity of just 0.16–0.98% of B . WF misses at most one feasible opportunity per setting (C_2 in Table 2).

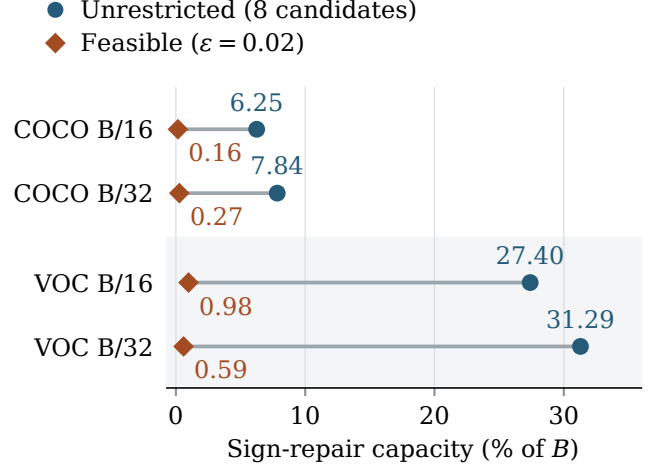


Figure 2: Budget-constrained sign-repair capacity among fixed screened failures B . Connected markers compare the same setting on a shared linear scale. Unrestricted: at least one of all eight candidates has $P \geq 0$; feasible: at least one also meets $\epsilon = .02$. Percentages use each setting’s own $|B|$ (Table 2) and are computed by exact enumeration.

Table 3: Budget endpoints (C: COCO; V: VOC). Multi: percentage of all eligible images with multiple feasible candidates. O_B, r_B : oracle/actual sign-repair percentages of screened failures B . $\Delta d_t, \Delta b$: all-image raw-target and bbox changes (bbox in percentage points).

Set	ϵ	Multi	O_B	r_B	Δd_t	Δb
C16	0.02	4.71	0.16	0.16	+0.0001	-0.12
C16	0.20	35.78	1.57	1.50	-0.0140	-1.63
C32	0.02	4.40	0.27	0.26	-0.0001	-0.08
C32	0.20	34.96	2.05	2.00	-0.0143	-0.60
V16	0.02	4.79	0.98	0.98	-0.0008	-0.34
V16	0.20	34.48	6.65	6.07	-0.0223	+0.08
V32	0.02	4.73	0.59	0.40	-0.0007	+0.02
V32	0.20	34.59	7.13	6.73	-0.0164	+1.11

Budget relaxation helps in every setting. At .02, 95.21–95.60% of images have only one feasible candidate (Table 3); WF has little room to change the region. At .20, oracle sign-repair capacity rises to 1.57–2.05% of COCO B and 6.65–7.13% of VOC B . Raw-target changes become negative in all four settings, but bbox changes are negative on COCO and positive on VOC. Budget relaxation therefore has dataset-dependent locality effects. Each comparison uses the same checkpoint and original population.

Population-wide and sign-repair effects. Table 3 includes unchanged regions. In contrast, Table 5 conditions on actual sign repairs within B ; The two tables describe population-wide changes and changes among successful sign repairs, respectively.

Table 4: Mean aggregate-margin differences over all eligible images, WF minus each alternative, with paired 95% CIs. Values are multiplied by 10^3 ; D counts WF/Max-.1 selection disagreements at $\epsilon = .02$.

Setting	WF-Mean	WF-Max-.1	D
COCO B/16	12.461	0.245	84
	[10.879,14.138]	[0.128,0.374]	
COCO B/32	9.343	0.241	89
	[7.986,10.801]	[0.083,0.414]	
VOC B/16	4.480	0.507	5
	[2.318,7.054]	[0.031,1.111]	
VOC B/32	3.184	0.418	5
	[1.467,5.133]	[0.058,0.927]	

Small baseline gains arise from few disagreements. WF and Max-.1 differ on only 84/89/5/5 images (Table 4). Their agreement is over 99.6% overall but 92.70–94.62% among images with multiple feasible candidates. On the 84 COCO B/16 disagreements, WF gains .0807 margin [.0443, .1171], while raw target drop changes by $-.0326$ [$-.0628$, $-.0033$]. The whole-sample gain is .000245. The VOC disagreement subsets contain only five images each.

Table 5: Actual sign repairs *within original B*, $\epsilon = .02$. r counts repairs; j counts those also satisfying J . WF-CCI means with repair-subset 95% CIs. Δd_t is raw held-out target-score change; only bbox change is in percentage points.

Setting	r	j	Δd_t	100 Δ bbox
COCO B/16	18	12	0.029	-19.53
			[-0.095,0.148]	[-38.61,-1.17]
COCO B/32	30	25	0.094	-14.04
			[0.037,0.149]	[-27.61,-1.48]
VOC B/16	5	3	-0.021	-16.87
			[-0.170,0.182]	[-26.11,-7.00]
VOC B/32	2	1	-0.078	-28.17
			[-0.115,-0.042]	[-78.57,22.22]

Target and locality changes after sign repair. In every setting, some sign repairs fail to retain the locality/mean-foil screen and positive target contribution ($j < r$ in Table 5). The joint feasible-oracle counts are 12/26/3/2. VOC estimates are based on small sign-repair subsets. Outside the screened population, sign transitions can also regress: COCO B/32 has four opposite-sign regressions among all baseline passes. Counts over all failures or all switched regions have different denominators from the screened sign repairs reported here.

Table 6: Nonzero-threshold sensitivity on fixed original A . Entries are percentages of screen-passers A with $P < -\delta$ using the normalized aggregate margin. Thresholds are evaluated post hoc in logit units.

Setting	$\delta = 0$	.01	.10	.50
COCO B/16	64.37	63.94	59.49	36.05
COCO B/32	64.78	64.38	60.22	35.19
VOC B/16	42.27	41.44	36.23	13.73
VOC B/32	41.16	40.34	34.47	11.98

Failure persists at larger margin thresholds. At $\delta = .10$, 59.49–60.22% of COCO and 34.47–36.23% of VOC screen-passers A still fail (Table 6). Rates fall further at .50, so the magnitude and threshold matter. The threshold sweep characterizes the magnitude of failure within the normalized aggregate score.

5 INTERPRETATION AND LIMITATIONS

Interpretation. The audit identifies a persistent gap between locality/mean-foil screening and worst-foil specificity. Matched-cardinality controls separate annotation exclusion from foil-count reduction. Candidate enumeration attributes limited sign-repair capacity to candidate availability and the target-drop budget, with few additional opportunities missed by WF.

Scope. The results concern CCI-top1, eight candidates, and the specified attention intervention. COCO/VOC differences combine foil count, target selection, and annotation conditions. Other candidate constructions and area-matched controls require further experiments. Bbox precision measures patch-center overlap. Semantic interpretation of hard foils requires annotation review and human validation. Small VOC sign-repair subsets limit precision of conditional endpoint estimates.

Reproducibility and normalization. Stored summaries and scripts support the threshold and budget reanalyses. Original masks and complete per-class responses are unavailable; region identity and normalization-induced changes in foil rankings remain unverified. Region-fixed comparisons assume faithful regeneration. Prompt holdout covers templates within the same images and models; cross-checkpoint budget calibration requires further evaluation.

6 CONCLUSION

Locality and positive mean-foil contrast leave substantial worst-foil failure in CCI-top1 regions, persisting after annotation exclusion and at larger margin thresholds. Candidate availability and target-drop constraints account for most limits on sign repair. Larger budgets increase capacity with setting-dependent target and locality effects. These findings support reporting worst-foil specificity alongside locality and mean-foil contrast when evaluating frozen CLIP region explanations.

REFERENCES

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 8748–8763. [Online]. Available: <https://proceedings.mlr.press/v139/radford21a.html>.
- [2] A. Agarwal, S. Karanam, and V. Gandhi, "Concept regions matter: Benchmarking CLIP with a new cluster-importance approach," *arXiv preprint arXiv:2511.12978*, 2025. [Online]. Available: <https://arxiv.org/abs/2511.12978>.
- [3] C. Lin, H. Chen, C. Kim, and S.-I. Lee, "Contrastive corpus attribution for explaining representations," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: <https://arxiv.org/abs/2210.00107>.
- [4] D. Williamson, Y. Ji, and M. Dwyer, "CASE: Contrastive activation for saliency estimation," *arXiv preprint arXiv:2506.07327*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.07327>.
- [5] Y. Wang, T. G. J. Rudner, and A. G. Wilson, "Visual explanations of image-text representations via multi-modal information bottleneck attribution," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 16 009–16 027. [Online]. Available: <https://doi.org/10.52202/075280-0704>.
- [6] Y. Li, H. Wang, Y. Duan, J. Zhang, and X. Li, "A closer look at the explainability of contrastive language-image pre-training," *Pattern Recognition*, vol. 162, p. 111409, 2025. [Online]. Available: <https://doi.org/10.1016/j.patcog.2025.111409>.
- [7] M. Dreyer, L. Hufe, J. Berend, T. Wiegand, S. Lapuschkin, and W. Samek, "From what to how: Attributing CLIP's latent components reveals unexpected semantic reliance," *arXiv preprint arXiv:2505.20229*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.20229>.
- [8] F. Li, Z. Meng, C. Tan, and Y. Yin, "Class-disentangled attribution with gradient guidance for CLIP," *Journal of King Saud University Computer and Information Sciences*, vol. 38, p. 376, 2026. [Online]. Available: <https://doi.org/10.1007/s44443-026-00779-3>.
- [9] M. Yuksekgonul, F. Bianchi, P. Kalluri, D. Jurafsky, and J. Zou, "When and why vision-language models behave like bags-of-words, and what to do about it?" in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: <https://openreview.net/forum?id=KRLUvvh8uaX>.
- [10] Z. Ye, F. Jiang, Q. Wang, K. Huang, and J. Huang, "IDEA: Image description enhanced CLIP-adapter for image classification," *Pattern Recognition*, vol. 171, p. 112224, 2026. [Online]. Available: <https://doi.org/10.1016/j.patcog.2025.112224>.
- [11] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollar, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Computer Vision—ECCV 2014*, ser. Lecture Notes in Computer Science, vol. 8693. Springer, 2014, pp. 740–755. [Online]. Available: https://doi.org/10.1007/978-3-319-10602-1_48.
- [12] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL visual object classes (VOC) challenge," *International Journal of Computer Vision*, vol. 88, no. 2, pp. 303–338, 2010. [Online]. Available: <https://doi.org/10.1007/s11263-009-0275-4>.
- [13] T. Gomez, T. Freour, and H. Mouchere, "Metrics for saliency map evaluation of deep learning explanation methods," in *Pattern Recognition and Artificial Intelligence: Third International Conference, ICPRAI 2022*, ser. Lecture Notes in Computer Science. Springer, 2022, pp. 84–95. [Online]. Available: https://doi.org/10.1007/978-3-031-09037-0_8.

COMPLIANCE WITH ETHICAL STANDARDS

This work analyses publicly released COCO and PASCAL VOC images and annotations. No participants were recruited and no new human or animal experiments were conducted.